UIDT / DATA EVIDENCE BRIEF

Weather and calendar effects

What the attached data can tell us - and how to use it

Reference dataset | 4 Bangalore junctions | July-September 2026

CURRENT EVIDENCE STATUS

Feature data is ready. Traffic impact is not yet measurable.

The file contains weather and calendar predictors, but no speed, flow, travel-time, occupancy, or congestion outcome.

57,008	3,631	77.7%	4
raw records	unique junction-hours	repeated raw rows	junctions

NORMAL DAY VERSUS CALENDAR CHANGES

Condition	Rows	Dates	Rain rate	Avg precip.	Humidity	Rain vs normal
Normal day	3,311	83	45.4%	0.220	71.1	baseline
Public holiday	80	2	36.2%	0.060	69.6	-9.2 pp
Long weekend	160	4	6.2%	0.009	57.6	-39.2 pp
Holiday + long weekend	80	2	45.0%	0.314	68.6	-0.4 pp

The long-weekend-only group is much drier than the normal-day reference, so weather and calendar effects are confounded.

NORMAL DAY: DRY VERSUS RAINY

Normal-day subset	Rows	Share	Avg precip.	Humidity	Wind
Dry	1,807	54.6%	0.000	68.2	13.4
Rain	1,504	45.4%	0.484	74.5	15.0

READ THIS CORRECTLY

These metrics describe the conditions in the dataset. They do not prove that rain, holidays, or long weekends changed traffic. That conclusion requires BTP traffic outcomes joined to the same junction and timestamp.

UIDT / MODEL IMPROVEMENT PLAN

Turn feature differences into traffic evidence

Use the attached file as explanatory context, then test whether it improves held-out BTP predictions.

RECOMMENDED PIPELINE

01 DEDUPLICATE
One row per junction + timestamp

02 ALIGN BTP
Join speed, travel time, flow, or occupancy

03 BASELINE
Junction + hour + day of week

04 ADD FEATURES
Rain, intensity, holiday, long weekend, weather

05 VALIDATE
Date split; compare errors and effect sizes

METRICS TO EXTRACT AFTER BTP ALIGNMENT

Evidence	What to calculate	Decision rule
Data quality	paired sample count, excluded rows, missingness	Must be reported
Traffic accuracy	MAE, RMSE, MAPE, bias	Compare baseline vs expanded
Weather effect	rain coefficient and confidence interval	Stable across dates/junctions
Calendar effect	holiday and long-weekend coefficients	Do not overclaim with sparse dates
Generalisation	held-out date performance	Primary go/no-go criterion

MODEL DECISION

Keep these features only if they improve held-out BTP error.

Model A: junction + hour + day of week. Model B: Model A plus rain, precipitation intensity, public holiday, long weekend, temperature, humidity, and wind. Report the difference in MAE/RMSE/MAPE with confidence intervals. Do not call the system validated until the BTP pairing is frozen and reproducible.

BOTTOM LINE

The attached data is valuable feature context. Its contribution to UIDT should be judged by measurable improvement on held-out BTP traffic outcomes, not by weather or holiday distributions alone.